Day-Old Chick Sexing using Convolutional Neural Network (CNN) and Computer Vision

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***Abstract*— As the Research and Development (R&D) Center of Zamboanga Peninsula (ZAMPEN) native chicken in the Philippines, the center supplies ZAMPEN native chicken to growers within the peninsula and its nearby regions. Most growers prefer to buy day-old chicks because of less handling and transfer costs. However, the accuracy rate of R&D staff in classifying day-old chicks thru feather sexing has an average of 55%. Because of this, the center does not sell day-old chicks to growers. This research aims to develop a device that classifies the sex of a day-old ZAMPEN native chick using CNN and computer vision through feather sexing. Two hundred (200) feather images were collected, consisting of 100 females and 1000 males. And 20 no-detection images were added to the data sets. The model was created by sequentially adding layers using VGG-16 architecture. The prototype captures a video of the day-old chick's feathers. Then the video frame is cropped and processed for classification. Thirty (30) day-old chicks were used to test the prototype and got an 83.33% accuracy.**

Keywords*— feather sexing, convolutional neural network, VGG-16, OpenCV, Raspberry Pi

I. Introduction

Over the past years, there has been an increase in the production of Philippine native chicken because of the rising number of health-conscious people in the country. Across the country, several native chicken breeds are raised. In the Zamboanga Peninsula (ZAMPEN), Joloano native chicken is commonly raised for cockfighting. In 2013, Western Mindanao State University (WMSU) and the Department of Science and Technology – Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development (DOST-PCAARD) conducted research that purifies the breed of Joloano chicken, which is later called ZAMPEN native chicken. The center's mandate is to help ensure food security and generate jobs in the Zamboanga Peninsula. The center aims to enhance breeding, feeding, and production management strategies to achieve its mandate. Here are the stages of growth of native chickens in the Center:

1. Brooding Stage – Week 1-3 or 21 days
2. Hardening Stage – Week 5-4 plus 6 days
3. Hardened Stage – Week 6-7 plus 6 days
4. Growing Stage – Week 8-15 plus 6 days
5. Pullet Cockerel – Week 16-20 plus 6 days
6. RTL – Week 21-23
7. Breeder – Week 24 and up

In the Brooding Stage, the chicks are separated by batch into male and female. In identifying the sex of day-old chicks, as shown in Fig. 1, the center uses feather sexing like Fig. 2 to 4 and should be performed by an expert. Chick wing feather patterns, which differ in sex due to genetic encoding, have been widely used for chick sex identification [1]. In feather sexing, the wings are spread to examine the primary and covert feathers, as shown in Fig. 2. Furthermore, the Aviagen Group [2] explained if the primary feathers are longer than the covert feathers, the chick is female, as shown in Fig. 3. If the coverts and primaries are the same lengths or shorter than the coverts, the chick is male, as shown in Fig. 4. The average accuracy rate of center's feather sexing is 55%.

Fig. 1. ZAMPEN day-old chick

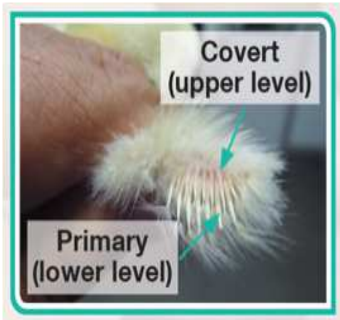

Fig. 2. Wing of a day-old broiler chick

Fig. 3. Female wing

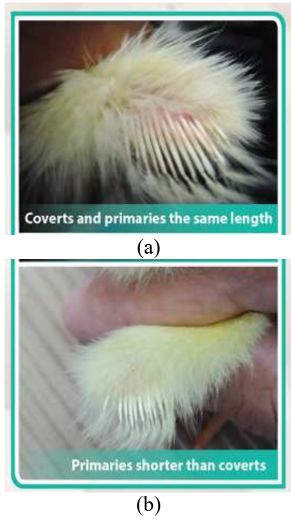

Fig. 4. Male wing type 1 (a) type 2 (b)

As computer vision and deep learning advance in their applications, it has become promising tool in the real-time automation of poultry monitoring systems due to its non-intrusive and non-invasive properties and ability to present a wide range of information [3]. However, there is little published information on computer vision in automating the sex determination of avian species [4]. But several researchers used Convolutional Neural Network (CNN) in identifying birds. The study of [5] used deep learning to model their suit of needs, which can be used to analyze the bird image. Moreover, PakhiChini used a deep learning model, i.e., CNN, that identifies individual birds from an input image [6].

This research also used tools used by other researchers that employed CNN and computer vision. CNN is a deep neural network specialized for image recognition, and it conquered most computer vision fields and is growing rapidly [7]. The device created [8] used Convolutional Neural Network-VGGNet-16 architecture to develop a device that identifies and detects abaca diseases such as ABTV and Mosaic, where the system achieved an 88.9% accuracy rate and an average precision rate of 91.7%. The Mobile application of [9] identified Bird Species using Tensorflow Hub pre-trained models to apply as a base model to their CNN network and compared the accuracy of models depending on their performance on two optimizers, Adam and RMSprop. The research of [10], pre-processing, classification, prediction, and creation are based on the TensorFlow toolkits and TensorFlow Keras library, which is applied to import the model and layer features utilized in pre-processing the dataset and modeling a CNN model and import other libraries for specific purposes like visualizing the dataset to understand the process further. OpenCV-Python is a real-time image processing and is considered the most popular infrastructure for computer vision applications [11]. [12] used the VGG-16 network as their model to extract the features from bird images. The system of [13] utilizes Python language, Pytorch model, and Raspberry Pi to perform this bird classification. And [14] used Raspberry Pi in their surveillance system for the farm from birds.

This research aims to develop a device that classifies the sex of a day-old ZAMPEN native chick using CNN and computer vision. It specifically aims to (1) build datasets of day-old ZAMPEN native chicks since there are no available datasets, (2) develop a model that classifies the sex of day-old chicks thru their wings, (3) use SOC as the main processor, and (4) use confusion matrix to validate the accuracy rate of the device.

The study is significant because it will help the center and other breeders of ZAMPEN native chicken to determine the sex of day-old chicks. Growers are now guaranteed the sex of the chicks they bought from the breeders. With this, the selling of brooder stock is faster and more reliable. Moreover, it also helps the center tag the chicks for breeding purification.

This research focuses on classifying ZAMPEN native day-old chicks using the feather sexing method. They will be classified through computer vision and CNN. The prototype can be only applied to ZAMPEN native day-old chicks excluding off-type.

II. Materials and Methods

In designing the prototype, Fig. 5 served as a guide. A 1080p USB camera will capture the feather of a day-old chick, focusing on its primaries and covert through computer vision. The captured video is converted into frames and then processed by the model.

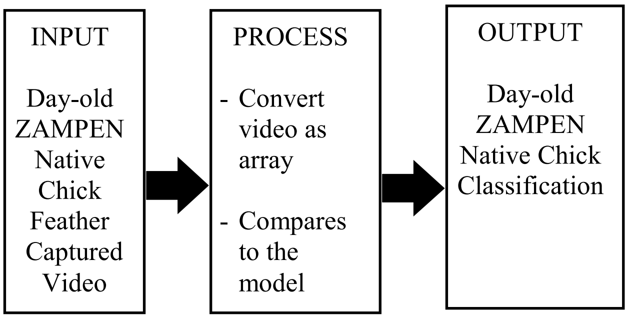

Fig. 5. Conceptual framework

A. Data Collection

The day-old chicks are manually identified by Rizalina and Archie, WMSU NICER R&D Center staff, through feather sexing introduced by Associate Professor Jocelyn Cuardra. Using a smartphone, 20-30 burst shot images focus on the feather's covert and primary parts. Then the chicks are tagged using numbered colored cable ties, as shown in Fig. 6.

Fig. 6. (a) Capturing and assigning of tags of 2nd batch (b) Green tag for Female day-old chicks (1st batch) (c) Blue tag for Male day-old chicks (1st batch)

After 21 days, they are physically checked, and cable ties are changed since their legs are getting bigger. Then after 45-50 days, it will be manually verified if the manual feather sexing on day-old chicks is accurate, as shown in Fig. 7. If incorrect, the images will be relabeled, and the model will be retrained. Five hundred (500) feather images were captured, but only 200 images were used in data sets due to mortality, 100 female and 100 male day-old chicks.

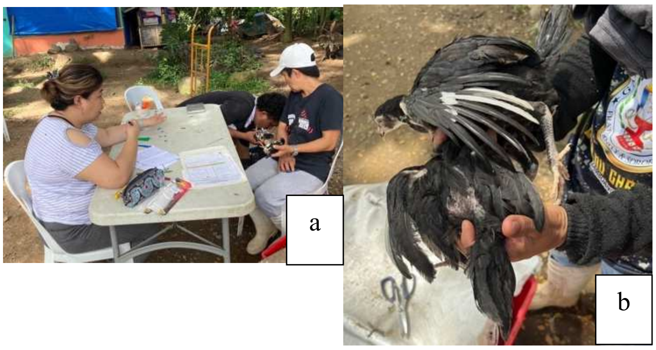

Fig. 7. (a) Retagging after 21 days (about 3 weeks) of the 1st batch (b) verification of sex after 45 days (about 1 and a half months) of batched 1

B. *Software Development*

In using CNN, the initial and most important step before feeding a validation image is making sure that the dimension of the image is the same as the requirement of the CNN [15]. The captured images are cropped, focusing on the wings to localize the feature, which is concentrated for analysis.

The images are labeled using LabelMe an open-source image annotation tool that allows users to label images. The collected images are annotated based on the chick feather structure, which has a shorter feather structure than the second for males and a longer feather structure than the second for females as shown in Fig. 8.

Fig 8. Sample feathers of ZAMPEN day-old chicks

Then Albumentations, an open-source Python library for image data augmentation was used. Data augmentation, segmentation, and sharpening were applied in image pre-processing to enhance image quality [16]. A sample of the augmented data is shown in Fig. 9. The image count after augmentation is as follows: 360 for test, 2880 for train, and 360 for validation.

Fig 9. Augmented male (left) and augmented female (Right)

After pre-processing the images, the images were loaded from the directory using TensorFlow dataset API. The loaded data were converted to a multi-dimensional array of elements of the same data type. It is implemented in the NumPy library for Python with a batch size of 32 and image size of 256 colors. The array will have an index for a label with elements of 0 and 1 representing the images' labels, 1 being Female and 0 for Male.

The model was created by sequentially adding the following layers using VGG-16, a pre-trained model. VGG-16 is one of the algorithms that achieved a high success rate in the ImageNet database [17]. The model was trained for 10 epochs (iterations over the entire training dataset). In each epoch, the model has processed 75 steps, and each step has a batch of data. Using OpenCV to read the image being captured and converting the image to RGB format with the resized method using TensorFlow. Keras load_model is used to load the created and pass the converted image to predict the loaded model giving the condition:

Threshold < 0.5 the image is MALE and Threshold > 0.5 the image is FEMALE.

C. *Prototype Development*

The design of the prototype is like an overhead projector where the 1080p USB camera is on top (bottom of the screen display) as shown in Fig. 10 side view, and at the bottom, the Raspberry Pi 3 Model B. The machine learning code for determining the gender of the chicks using video feedback and wing analysis was deployed on Raspberry Pi 3 Model B+. It is equipped with Broadcom BCM2837 64bit CPU with a clock speed of 1.4GHz. The attached camera should have a high resolution as the recommendation by [18], for more accurate results and proper illumination in the imaging chamber. For the user to be informed of the status classification, an RPi LCD screen is attached.

D. *Prototype Testing*

For the device can classify the sex of the day-old chick, the user needs to spread the feathers and place it on the target area and steady for 5-10 seconds. Based on the captured video, it is compared to the trained model for classification. When the classification is done, the user is informed through LED indicators and screen display, as shown in Fig. 10.

Fig10. Day-old chick sexing prototype

The prototype takes a video of the feathers of a day-old chick. Before processing, the video is cropped and then loaded into the model for classification. The result of the classification is displayed through LED and screen. The detailed flow is shown in Fig. 11.

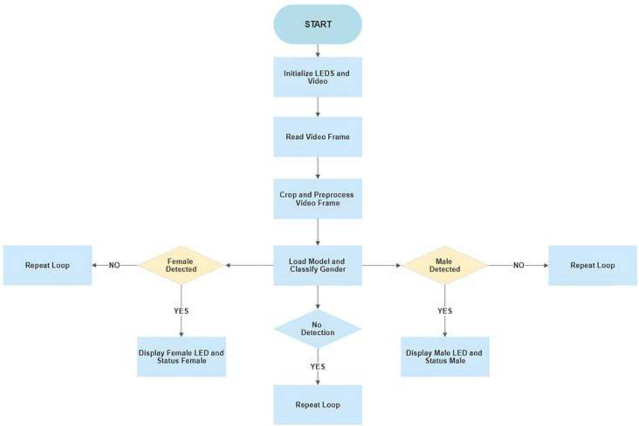

Fig 11. Flowchart of the prototype

III. Result and Discussion

There were sixty-eight (68) day-old chicks used for testing but after 45 days (about 1 and a half months) only 30 chicks survived. The result of the classification is shown in Table I.

TABLE I. TESTING DATA

Sample	**Sex**	
	21 days*	***45 days*****
1	Female	Male
2	Female	Male
3	Female	Female
4	Female	Female
5	Female	Female
6	Female	Female
7	Female	Female
8	Female	Female
9	Female	Female
10	Female	Female
11	Female	Female
12	Female	Female
13	Female	Female
14	Female	Female
15	Female	Female
16	Female	Female
17	Male	Female
18	Male	Female
19	Male	Male
20	Male	Female
21	Male	Male
22	Male	Male
23	Male	Male
24	Male	Male
25	Male	Male
26	Male	Male
27	Male	Male
28	Male	Male
29	Male	Male
30	Male	Male

Legend:
* classified by the prototype
** classified by human

The accuracy of the prototype is determined using confusion matric as shown in Table II.

TABLE II. CONFUSION MATRIX

	Predicted			
Actual	***Total Sample = 30***	***Female***	***Male***	***Total Actual***
	Female	14	3	17
	Male	2	11	13
	Total Predicted	16	14	30

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} x100 \qquad (1)$$

TABLE III. ACCURACY FOR EACH CLASSIFIED SEX

Classification	**Accuracy**
Female	87.50
Male	78.57

Using equation (1), the device's accuracy is 83.33% which is 28.3% higher than the R&D staff. The device's accuracy in classifying sex is higher on female day-old chicks as shown in Table III where the R&D staff have less accuracy.

IV. Conclusion and Future Works

After several tests and recalibration, the researcher developed a prototype with a higher classification accuracy than the R&D staff. With a model of 220 images with the aid of CNN, VGG-16, Open CV, and TensorFlow, the prototype gained an accuracy rate of 83.33%. And it has a higher accuracy rate on the female classification of 87.50% while 78.57% for male. The training data consists of certain characteristics that are more common in female chicks, the model struggles to generalize to other types of chicks, leading to lower accuracy for males.

This study recommends increasing the images of the datasets to achieve a higher accuracy rate.

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